



These synaptic transistors can sense pain

A new type of artificial skin was created by a team of engineers from the University of Glasgow using a novel processing system based on 'synaptic transistors,' which replicates the brain's neural networks in order to learn. The researchers printed a grid of 168 synaptic transistors

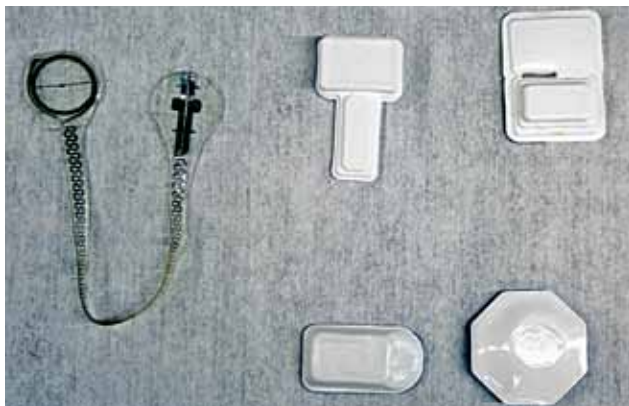


The robot hand used for the experiment (Credit: University of Glasgow)

manufactured from zinc-oxide nanowires directly onto the surface of a flexible plastic to create an electronic skin capable of a computationally efficient, synapse-like response. It was then attached to a robot hand. The team trained the skin to respond appropriately to simulated pain using the varied voltages, which then triggered the robot hand to react. It actually learned to move away from the source of discomfort!

A pacemaker that can dissolve in the body after use

Researchers from Northwestern University have unveiled a new, smart pacemaker that is connected to a network of

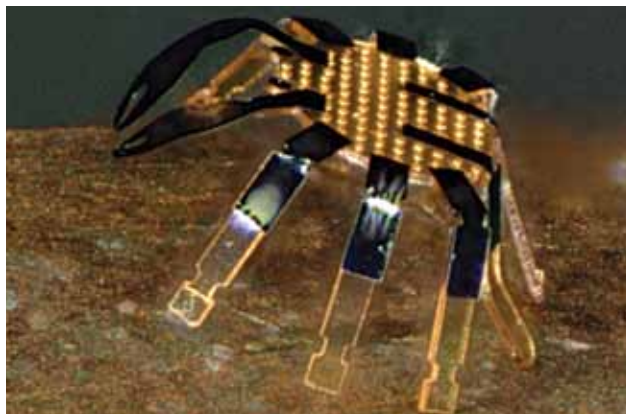


All devices laid out together, with the transient pacemaker on the left (Credit: Northwestern University)

soft, flexible, wireless sensors and control units. The multiple sensors present in the system communicate with one another to keep track of the body's numerous physiological activities, such as body temperature, oxygen level, respiration, muscle tone, physical activity, and the electrical activity of the heart. All of this data is sent to a smartphone or tablet, allowing doctors to keep an eye on their patients from afar. After a time of use, the pacemaker degrades naturally in the body and distributes an anti-inflammatory medication to avoid foreign-body reactions as it dissolves slowly and painlessly.

Engineers develop a tiny remote-controlled robot crab

Northwestern University engineers have unveiled the world's tiniest remote-controlled walking robot—a tiny robotic crab! The researchers believe their device is a



A magnified view of the tiny crab robot, standing on the edge of a coin (Credit: Northwestern University)

significant step toward developing practical robots that can execute remote-controlled operations in small settings. It comprises a shape-memory alloy, which allows it to change shape when heated before returning to its original shape when cooled. The team was able to heat specific portions of the robot using a laser beam, forcing it to walk. Also, because these structures are so tiny, the rate of cooling is very fast. This bot certainly paves the way for more advanced micro-robots in the future.

Bioinspired vision sensors that mimic human vision

Researchers at Hong Kong Polytechnic University (PolyU) and Yonsei University in Seoul have developed vision sensors that mimic and even surpass the ability of the human retina to adjust to different lighting levels. Each of the new vision sensors is made up of arrays of phototransistors.